README

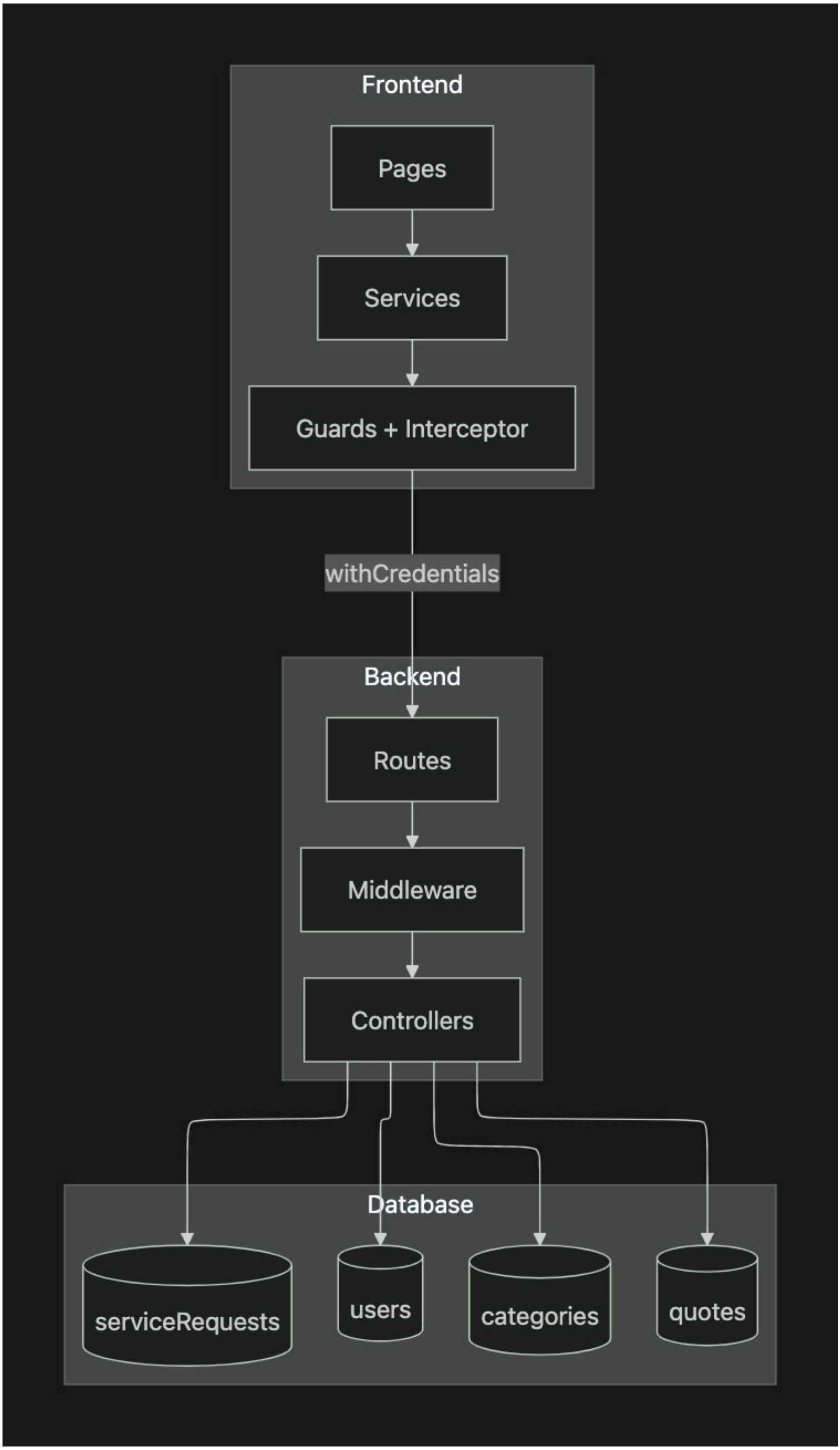
Neighborhood Service Marketplace (NSM)

Full-stack web application where Residents post service requests and Providers submit quotes. Built with MongoDB, Express, Angular, and Node.js. Session-based authentication with role-based access control.

Team

Name	Student Number	Section
Bessong Ekep Obasi Arrey-Etta	n01708502	ON-A
Raymond Choy	n01743348	ON-A
Harshit Sahni	n01734894	ON-A
Marcia Oliveira	n01716166	ON-A
Phuoc Huy Truong	n01741637	ON-B

System Architecture


Frontend
Pages
Services
Guards + Interceptor
withCredentials
Backend
Routes
Middleware
Controllers
Database
serviceRequests
users
categories
quotes

Backend —> Express server with session-based auth, role middleware, Mongoose ODM.
Frontend —> Angular standalone components, reactive forms, Tailwind CSS, HTTP services with credentials interceptor.

Folder Structure

```
backend/src/
  config/       db.js, session.js
  utils/        constants.js, env.js
  models/       users.js, Category.js, ServiceRequest.js, Quote.js
  controllers/  authController.js, categoryController.js, requestController.js, quoteC
ontroller.js
  middleware/   requireAuth.js, requireRole.js, errorHandler.js
  routes/       authRoutes.js, categoryRoutes.js, requestRoutes.js, quoteRoutes.js
  app.js

frontend/src/app/
  core/         services/ (auth, category, request, quote), guards/ (auth, role), inte
rceptors/
  models/       user, category, service-request, quote TypeScript interfaces
  pages/        login, register, requests-list, create-request, request-details, my-qu
otes
  shared/       navbar
```

Setup Instructions

Prerequisites

- Node.js 18+
- MongoDB (local, Docker, or Atlas connection string)

Quick Start

macOS / Linux:

```
chmod +x start.sh && ./start.sh
```

Windows:

```
start.bat
```

Manual Setup

Backend:

```
cd backend
cp .env.example .env    # edit MONGODB_URI if using Atlas
npm install
npm start               # runs on http://localhost:3000
```

Frontend:

```
cd frontend
npm install
ng serve                # runs on http://localhost:4200
```

Environment Variables

Variable	Default	Description
`MONGODB_URI`	`mongodb://localhost:27017/nsm`	MongoDB connection string

Variable	Default	Description
`SESSION_SECRET`	`devsecret`	Session encryption key
`CLIENT_ORIGIN`	`http://localhost:4200`	CORS allowed origin
`PORT`	`3000`	Backend server port

Database Schema

Four collections using a **referencing strategy** to avoid document bloat and keep updates simple.

users

Field	Type	Constraints
fullName	String	required, 2–80 chars
email	String	required, unique, lowercase
passwordHash	String	required, bcrypt hashed
role	String	enum: resident, provider

categories

Field	Type	Constraints
name	String	required, unique, 2–50 chars
description	String	optional, max 200 chars

serviceRequests

Field	Type	Constraints
title	String	required, 5–80 chars
description	String	required, 10–1000 chars
categoryId	ObjectId ref	required, references categories
createdBy	ObjectId ref	required, references users
location	String	required, 2–80 chars
status	String	enum: open, quoted, assigned, completed, cancelled
acceptedQuoteId	ObjectId ref	set when quote accepted

quotes

Field	Type	Constraints
requestId	ObjectId ref	required, references serviceRequests
providerId	ObjectId ref	required, references users
price	Number	required, min 1
message	String	required, 5–500 chars
daysToComplete	Number	required, 1–30
status	String	enum: pending, accepted, rejected

Schema Design Justification

Why referencing over embedding:

- A request can receive many quotes. Embedding quotes inside requests leads to unbounded array growth and complicates pagination.
- Users and categories are shared across many documents. Referencing prevents duplication and ensures a single source of truth for updates.
- Quotes need independent status tracking and lifecycle management. Separate documents allow targeted queries and atomic status updates.

Query Patterns

1. **List requests with filters,** filter by `status`, `categoryId`, keyword via `$text` search. Uses compound index `(status, categoryId)` and text index on `(title, description)`.
2. **Get quotes for a request,** lookup by `requestId`. Uses index on `quotes.requestId`.
3. **Provider "My Quotes",** lookup by `providerId`. Uses index on `quotes.providerId`.

Indexing Strategy

Index	Collection	Purpose
`{ email: 1 }` unique	users	Fast login lookup, prevent duplicates
`{ name: 1 }` unique	categories	Prevent duplicate category names
`{ title: "text", description: "text" }`	serviceRequests	Keyword search across request content
`{ status: 1, categoryId: 1 }`	serviceRequests	Compound filter for listing with status and category
`{ requestId: 1 }`	quotes	Fast lookup of all quotes for a request
`{ providerId: 1 }`	quotes	Fast lookup for provider's own quotes
`{ requestId: 1, providerId: 1 }` unique	quotes	Prevent duplicate quotes per provider per request

These indexes support the three primary query patterns. The compound unique index on quotes also enforces a business rule at the database level. The text index enables the keyword search requirement without additional infrastructure.

Scalability Considerations

- Referencing allows independent scaling of collections. Quotes can grow without affecting request document size.
- Indexes cover the primary access patterns. Additional indexes (e.g., on `createdBy` for "my requests" views) can be added as query patterns evolve.
- The accept-quote operation updates multiple documents sequentially. For production workloads with concurrent access, MongoDB transactions with a replica set would guarantee atomicity.
- Session storage uses in-memory store for development. Production deployments should use `connect-mongo` for persistent session storage.

API Endpoints

Auth

Method	Endpoint	Access	Description
POST	`/api/auth/register`	Public	Register user (fullName, email, password, role)
POST	`/api/auth/login`	Public	Login, set session cookie
POST	`/api/auth/logout`	Public	Destroy session
GET	`/api/auth/me`	Authenticated	Get current user profile

Categories

Method	Endpoint	Access	Description
GET	`/api/categories`	Public	List all categories
POST	`/api/categories`	Authenticated	Create category

Service Requests

Method	Endpoint	Access	Description
POST	`/api/requests`	Resident	Create service request
GET	`/api/requests`	Authenticated	List requests (filters: status, categoryId, q)
GET	`/api/requests/:id`	Authenticated	Get request details with populated refs
PATCH	`/api/requests/:id/status`	Resident (owner)	Update request status

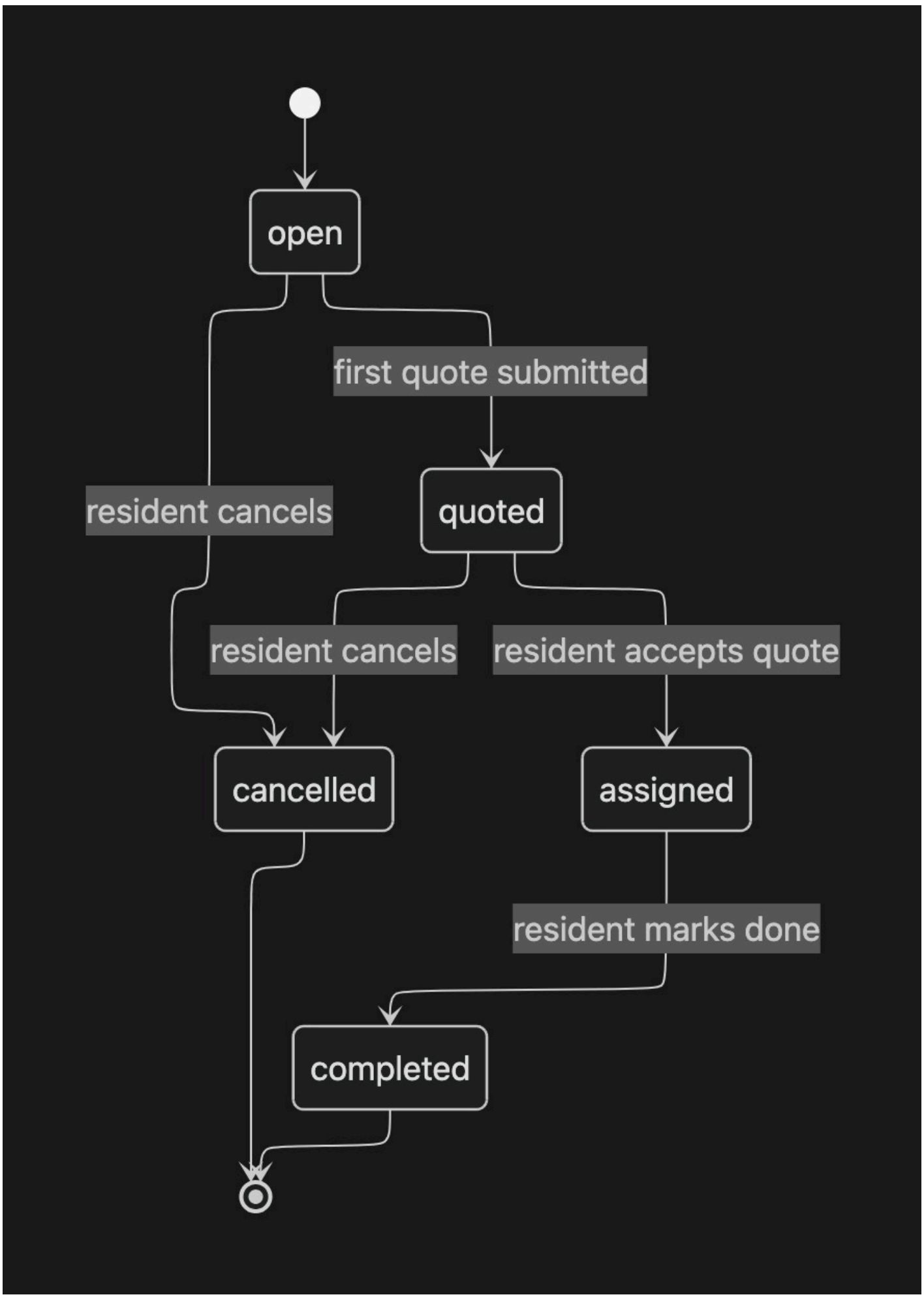
Quotes

Method	Endpoint	Access	Description
POST	`/api/requests/:id/quotes`	Provider	Submit quote for a request

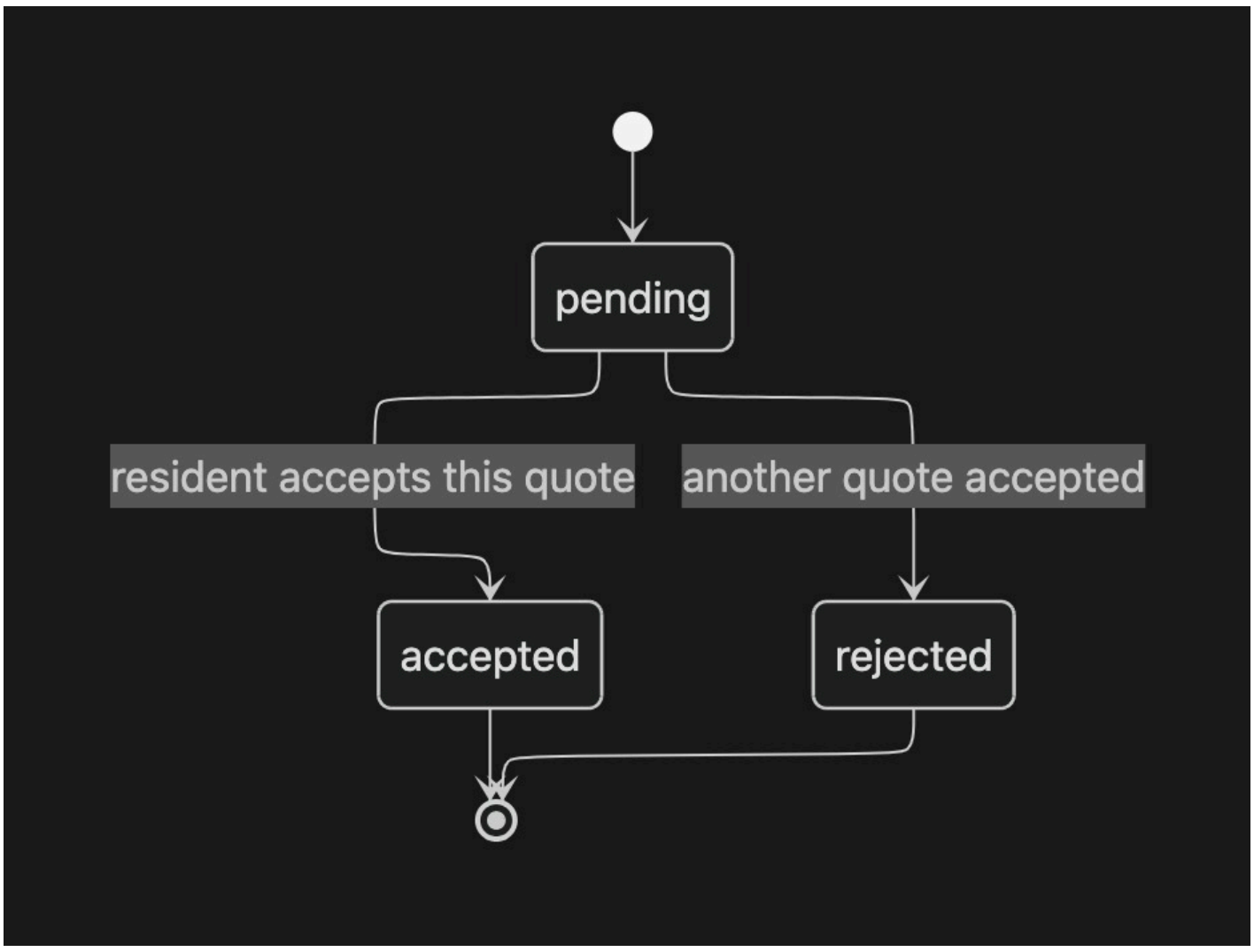
Method	Endpoint	Access	Description
GET	`/api/requests/:id/quotes`	Authenticated	Get quotes for a request
PATCH	`/api/quotes/:id/accept`	Resident (owner)	Accept quote, reject others, assign request
GET	`/api/quotes/mine`	Provider	Get provider's own quotes

Status Lifecycle

Request Statuses



Quote Statuses



Testing

Postman

Import `NSM.postman_collection.json` into Postman. Run requests in order (top to bottom). The collection auto-saves IDs between requests using collection variables.

Covered scenarios:

1. Register resident and provider
2. Duplicate email rejection (409)
3. Login resident
4. Create category
5. Create service request
6. Provider login and quote submission
7. Auto-transition to “quoted” status
8. Resident views and accepts quote
9. Verify assigned status and acceptedQuoteId
10. Mark request completed